DEPARTMENT OF DATA ANALYTICS & MATHEMATICAL SCIENCES

JAIN DEEMED-TO-BE UNIVERSITY

SYNOPSIS

Analyzing Academic and Behavioral Factors for Student Performance Prediction

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Current Status	Data Cleaning & EDA — Model Development (In Progress)

ABSTRACT

Student academic performance is influenced by many factors beyond marks and attendance. Behavioral and psychological factors such as stress, procrastination, lack of sleep, and poor time management have become major concerns. Most traditional academic systems focus only on examination results and fail to identify the hidden reasons behind poor performance.

This project develops an AI-based student failure prediction and early risk detection system using both academic and behavioral factors. Data was collected via a Google Form survey from 52 students across 29 fields covering study hours, attendance, sleep patterns, stress levels, procrastination habits, and CGPA.

The dataset was cleaned and preprocessed using Pandas and NumPy. Feature engineering created composite behavioral variables — Stress Score and Procrastination Score. EDA was performed to understand relationships between features and CGPA. The proposed system uses Linear Regression (baseline) alongside advanced ensemble models to predict performance and classify at-risk students.

INTRODUCTION

Academic performance prediction has emerged as a key area in educational data science. Most institutions evaluate students primarily on marks and attendance — surface-level indicators that fail to explain the behavioral and psychological root causes of poor performance.

Students face challenges including stress, anxiety, procrastination, irregular sleep, and poor time management. These hidden factors directly affect concentration and learning ability. In many cases, struggling students are only identified after examination failure — by which point timely intervention has been missed.

Machine learning can analyze patterns from student data and flag at-risk students earlier. By combining academic and behavioral features, institutions can better understand root causes of failure and provide targeted support. This project bridges behavioral psychology with educational data mining to create a meaningful, explainable prediction system.

LITERATURE SURVEY

M. Goyal & S. Kumar (2024) — *Early Detection of At-Risk Students Using Machine Learning [arXiv:2412.09483]*

Finding: ML models using academic data can predict at-risk students with up to 87% accuracy. **Gap:** Does not include stress or procrastination; limited behavioral analysis.

F. Rozental et al. (2022) — *Academic Procrastination in University Students: Systematic Review [PLoS One, vol. 17, no. 3]*

Finding: Academic procrastination has a significant negative effect on performance across disciplines. **Gap:** Purely psychological/statistical — no ML predictive model developed.

A. S. Dhall et al. (2023) — *Stress and Anxiety Level Prediction in Students Using ML [IEEE ICOSEC]*

Finding: ML techniques predict stress and anxiety levels with accuracy above 85% using survey features. **Gap:** Focuses only on stress/anxiety prediction; does not link to CGPA prediction.

P. Gupta & R. Pathak (2020) — *Multi-factor ML Framework for Predicting Student Academic Performance [PLoS One, vol. 15]*

Finding: A multi-factor ML framework improves academic performance prediction vs. single-factor models. **Gap:** Does not consider behavioral factors like stress and procrastination.

Summary: Most studies address either academic prediction or behavioral analysis separately. Very few combine stress, procrastination, and academic data in a single unified prediction system — the gap this project directly addresses.

RESEARCH GAP

- Most existing systems use only academic factors (marks, attendance) for prediction.
- Behavioral factors like stress and procrastination are studied in isolation from academic outcomes.
- Very limited work combines behavioral psychology with machine learning for academic prediction.
- Existing systems predict outcomes but do not explain the behavioral reasons behind poor performance.
- Early identification of at-risk postgraduate students using behavioral signals is largely unexplored.
- Stress and procrastination have never been jointly engineered as composite ML features for CGPA prediction.

PROBLEM STATEMENT

Students in higher education often face academic difficulties due to hidden behavioral and psychological factors — stress, procrastination, poor sleep, and irregular study routines. Traditional academic monitoring systems focus on marks and attendance and react only after poor performance has already occurred, providing no early warning and no explanation of the root cause. There is a clear need for a proactive system that analyzes both academic and behavioral factors, classifies students by risk level, and delivers interpretable insights to faculty for timely intervention.

OBJECTIVES

1. To collect and analyze behavioral and academic data from M.Sc. students using validated survey instruments.
2. To study the impact of stress and procrastination on academic performance (CGPA).
3. To preprocess, clean, and prepare the collected dataset for machine learning.
4. To create meaningful composite behavioral features via feature engineering (Stress Score, Procrastination Score).
5. To perform Exploratory Data Analysis (EDA) to understand data distributions and feature relationships.
6. To develop a machine learning-based prediction system using Linear Regression (baseline) and ensemble models.
7. To classify students into Low / Medium / High academic risk categories based on predicted CGPA thresholds.
8. To provide reason-based behavioral insights enabling targeted early intervention by faculty.
9. To deploy a Streamlit-based web application for real-time usage by educators.

DATA SOURCE & EXPLORATORY DATA ANALYSIS

Data Source

The dataset was collected via a Google Form survey (Student Academic Behavior Survey) distributed to M.Sc. Data Science & Analytics students at JAIN Deemed-to-be University.

Parameter	Details
Total Responses	52
Survey Fields	29 questions
Data Type	Primary (self-reported)
Final Dataset	52 rows × 12 numeric columns, 0 missing values
Key Variables	Study hours, Sleep hours, Attendance %, CGPA, Stress Score, Procrastination Score

Data Cleaning & Feature Engineering

- Standardization of column names to snake_case
- Removal of unnecessary columns (Timestamp, Username, Hobbies)
- Median imputation for 2 missing values
- Categorical survey ranges encoded to numeric midpoints
- Stress Score: composite of 5 stress-related survey questions
- Procrastination Score: composite of 10 procrastination-related questions

EDA — Key Findings

- Higher study hours correlate positively with CGPA (scatter slope = 0.16).
- Elevated stress is associated with lower CGPA; low-stress students show the most stable performance.
- Procrastination score shows a slight negative correlation with CGPA.
- Stress and procrastination scores are strongly correlated (r = 0.57), indicating co-occurring risks.
- Students with medium stress show highest CGPA variance, suggesting other moderating factors.
- Techniques used: histograms, scatter plots (feature vs. CGPA), correlation heatmaps, and boxplots.

METHODOLOGY

The project follows a structured seven-step pipeline combining data science and behavioral psychology:

Step	Phase	Description
1	Data Collection	Primary data collected via Google Form survey (52 responses, 29 fields).
2	Data Cleaning & Preprocessing	Column standardization, removal of irrelevant fields, median imputation, categorical encoding to numeric midpoints.
3	Feature Engineering	Composite Stress Score (5 questions) and Procrastination Score (10 questions) computed.
4	Exploratory Data Analysis	Distributions, scatter plots (feature vs. CGPA), heatmaps, and boxplots analyzed.
5	Data Preparation	Feature selection and train-test split (80:20) for model building.
6	Model Development	Linear Regression as baseline; Random Forest & Gradient Boosting planned. Evaluated via MAE, RMSE, and R^2.
7	Risk Classification & Deployment	Students classified into Low / Medium / High risk. Streamlit web app for real-time prediction.

CURRENT STATUS OF WORK

COMPLETED	PENDING / IN PROGRESS
• Literature survey completed	• ML model implementation (baseline: Linear Regression)
• Research gap identified	• Advanced models: Random Forest, Gradient Boosting
• Google Form survey designed and distributed	• Model evaluation (MAE, RMSE, R^2)
• Dataset collection completed (52 responses)	• SHAP explainability integration
• Data cleaning & preprocessing completed	• Dataset expansion (target: 500+ responses)
• Feature engineering completed	• Risk classification module
• Exploratory Data Analysis (EDA) completed	• Streamlit web application development & deployment

EXPECTED OUTCOME

The expected outcome is a fully functional AI-based early academic risk detection system that:

- Accurately predicts student CGPA from combined academic and behavioral features.
- Classifies students into Low / Medium / High academic risk categories.
- Identifies students with elevated stress and procrastination as early warning signals.
- Provides interpretable behavioral explanations for underperformance using SHAP values.
- Supports early academic intervention by faculty before exam failure occurs.
- Is accessible as a real-time Streamlit web application for institutional use.

ACTION PLAN

Phase 1: Dataset Expansion

- Collect 500+ responses across multiple programmes and batches.
- Distribute form to additional MSc and MBA cohorts.
- Validate expanded dataset quality and re-run preprocessing pipeline.

Phase 2: Advanced ML Models

- Implement Random Forest and Gradient Boosting alongside baseline.
- Compare models using MAE, RMSE, and R^2 metrics.
- Add risk classification layer (Low / Medium / High CGPA labels).

Phase 3: Explainability (SHAP)

- Integrate SHAP values to explain individual predictions.
- Generate per-student behavioral insight reports.
- Visualize feature contributions for faculty dashboards.

Phase 4: Web App Deployment

- Build Streamlit web application with real-time risk prediction.
- Deploy on Streamlit Cloud for institutional pilot testing.
- Conduct pilot test with faculty and incorporate feedback.

CONCLUSION

This project bridges educational data mining and behavioral psychology by combining academic and behavioral factors to predict student performance. Stress and procrastination are measurable, encodable, and predictive behavioral indicators that influence CGPA outcomes.

The 52-response survey dataset was successfully cleaned, feature-engineered, and analyzed through EDA. Key findings confirm that study hours correlate positively with CGPA, while elevated stress and procrastination act as negative risk signals — with stress and procrastination also strongly co-occurring ($r = 0.57$).

The project demonstrates how machine learning and behavioral analysis can be combined into a proactive, interpretable academic support system — moving beyond reactive grade-based evaluation

toward early, reason-based intervention. The dataset is fully prepared for model training, evaluation, and Streamlit deployment.

"Predictive AI is ultimately a tool for human empathy — by measuring what was previously invisible, we can build education systems that support the whole student."

REFERENCES

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